

# Isaac Mubiru

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## PROFESSIONAL SUMMARY

Senior full-stack engineer with 6+ years shipping product end to end — from React interfaces down to the Node.js/TypeScript services and async job pipelines behind them. Led the migration of 20+ production apps from Parse/MongoDB to Firebase at MTailor with zero downtime, cutting significant monthly infrastructure costs while keeping a live system running throughout, and built the real-time two-way Pub/Sub sync that made it possible. Also shipped the revenue-moving features on top: a 3D product visualisation that lifted buyer conversion and an Express Shipping flow worth \$40 per order. Four years fully remote with US and Singapore teams, reporting directly to CTOs. Based in Kampala, Uganda (UTC+3); written-first and comfortable owning ambiguous work across time zones.

## CORE COMPETENCIES

End-to-End Feature Ownership

React Product Interfaces

Node.js + TypeScript APIs

Async Job & Event Pipelines

Real-Time Data Sync

Microservices (Kafka, gRPC, Pub/Sub)

Docker & Kubernetes

Technical Writing & Enablement

## WORK EXPERIENCE

### MTailor

September 2022 – Present

Software Engineer (US Remote) — reporting to the CTO

- Led the **end-to-end migration of 20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, owning architecture through cutover
- Built a **3D visualisation feature** for customers using video overlay and the ffmpeg library — shipped from the UI down, **increasing buyer conversion**
- Implemented **Express Shipping** end to end (backend, pricing, checkout UI, ops handoff), generating **\$40 additional revenue per order**
- Implemented **real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub** — idempotent, redelivery-tolerant async job processing while both stores served live traffic
- Migrated file storage from Amazon S3 to Google Cloud Storage via a Python migration script
- Migrated a Webflow marketing site from a Parse backend to Firebase
- Saved the company **\$5,000/month** by migrating all services off AWS

### MTailor

March 2022 – July 2022

Contractor (US Remote) — Upwork-sourced backend migration

- Built a migration tool to read from MongoDB and write to Firestore after complex transformation logic
- Wrote the documentation fellow engineers onboarded from for the new Firebase SDKs (Node.js and Python)
- Trained the Ops team on the new Firebase dashboard and backup procedures

### Dr Wealth

August 2021 – January 2022

Software Engineer (Singapore Remote)

- Led optimisation of a **PWA delivering real-time stock market prices** — a live-updating, data-dense product surface on a Firebase backend
- Added responsive UI pages to the PWA using React, Tailwind, HTML and CSS
- Built ad-hoc jobs to **query over 2 million Firestore records**, keeping customer prices current from Morningstar APIs
- Extended backend APIs using **Firebase Cloud Functions and Express.js** hosted on Heroku

### Mind2matter

January 2022 – July 2022

Software Engineer (US Remote) — reporting to the CTO

- Delivered React UIs from Figma designs** under tight timelines for demanding agency clients
- Built backends for DeFi applications using Web3 and Node.js

### CodeBits

January 2020 – July 2021

Software Engineer / Team Lead (Uganda Remote)

- Led a team of 4 developers** building legal-tech systems for NGOs in Uganda
- Built a **gRPC service contract** that external legal-aid providers integrated against, backed by a Node.js service and a USSD gateway
- Architected a microservices backend** for FIDA Uganda using Apache Kafka, NATS Streaming, Docker and Kubernetes
- Built a case management system for FIDA Uganda and a paralegal database for LASPNET Uganda

- Developed a cross-platform mobile app for LASPNET using React Native and Expo, with Firebase push notifications

## SELECTED PROJECTS

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### career-ops

[Open Source](#)

Agentic job-search pipeline built on Node.js and Claude Code: live URL verification with Playwright, YAML-configured scanning, automated PDF generation. Used to evaluate 100+ roles end to end.

Node.js · Playwright · YAML · CLI tooling

### Healthcare WhatsApp Chatbot

Conversational product integrating OpenAI and Google Vision for prescription OCR, delivered over the WhatsApp Business API.

Node.js · OpenAI · Google Vision · WhatsApp Business API

## EDUCATION

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**B.Sc. Computer Engineering** · [Makerere University, Kampala](#)

2018 – 2022

## SKILLS

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**Proficient:** Node.js, TypeScript, JavaScript, React, Firebase, GCP (Firestore, Pub/Sub, Cloud Functions, GCS), MongoDB, SQL, Python, HTML, CSS, Tailwind, Git, Linux

**Intermediate:** Go, Rust, AWS (S3, EC2, EBS), React Native, Docker, Kubernetes, Kafka, NATS, gRPC

**Languages:** English, Kiswahili